

can be used to generate and supply extra power to the grids in the morning and then consume the power at night if it needs it," says Prabaljeet Kalia, product manager - drives and motion, Fuji Electric.

Most energy systems (oil, wind and solar) from the last 30 years have been installed as open-loop systems, creating a challenging business case for sustained business feasibility. Technology is now enabling these energy verticals to be integrated into a holistic ecosystem, linking these together in a closed-loop structure. "As an example, multiple residual constituents (such as natural gas or carbon-dioxide) are by-products of energy extraction. Technology is enabling these constituents to be used as renewable sources of energy that can be used in a closed-loop environment," explains Shuey.

Embitel, a product engineering services company based in Bengaluru, is connecting the elements of the energy ecosystem through technologies like sensors, communication gateways, cloud storage and retrieval, data analytics and finally the graphical user interfaces (GUIs) that enable operators to make real-time decisions to optimise the performance of the energy domain. These are all essential to enabling an energy system that meets the needs of the global community.

Alok Chandna, power and social infrastructure initiative, Fuji Electric, says, "We have partnered with a company named Lotus Wireless that provides management and control systems for developing building energy management systems (BEMSes). BEMS depends a lot on weather conditions. For instance, if you know that tomorrow will not be sunny, you know that your solar-energy-harvesting system will not work, so you will need to charge your batteries in advance so that you can use these or figure out an alternative source of energy like wind or hydro or a backup grid."

BEMS is the basic building block of forming a community energy-man-

agement systems (CEMSes) in capital, which, in turn, leads to forming micro-grids. The benefit here is that even if a natural disaster were to hit and all sub-stations and power generation was lost, a micro-grid would be able to bring the city back to life. In a way, this can be seen as an aid to disaster management as well. Fuji Electric Japan has already partnered with a state government in India to implement micro-grids.

Today there are engineers who are considering using ultracapacitors instead of batteries. Why? The tipping point that caused people to move from batteries to ultracapacitors is driven by battery maintenance and replacement cost. Lifetime cost is also lower since you have to change batteries frequently; ultracapacitors last much longer.

Grids

The problem is that, we also see almost 40 per cent transmission and distribution losses in our grids. "This means you need to make it efficient first and then make it smart. Micro-grids are something that I am very bullish on. A concept like this where the community can drive their own self-sustaining power is absolutely fantastic," says Sanjay Kumar, CEO and MD of Altran India.

Shuey says that firms like Embitel Technologies are building connections and partnerships with leading clean-tech companies that are providing solutions to the energy sector. "We are working with companies delivering real-time monitoring solutions using the latest sensing technologies, companies that are delivering power-generation systems that run on alternative fuel and those that enable closed-loop systems that link traditional fossil fuel and renewable-energy solutions. It all starts with collaboration and partnership; demonstrating solutions where there is the largest return on investment. The days of junk science are over."

The idea is to start from the consumer side by implementing

smartmeters that have two-way communication, as well as the ability for the utility company to connect or disconnect the power supply itself. "It also tracks usage data and status to provide the utility company with data to help make its system more efficient. Outage management systems are also implemented these days in tandem with Fuji Electric's SCADA system to provide information to call centres and consumers to get information on status of grids, reasons for failure and estimated time for it to be back in action," says Chandna. Fuji Electric intends to provide a Haryana State Electricity Distribution company an end-to-end solution including smart metering, meter data management and outage management.

What is driving this sector? Decentralised solar power plants, new energy-storage solutions, higher-efficiency inverters, which can deliver efficiency to the tune of 98.3 per cent and better cooling systems have come up. "The rooftop market is what excited me the most. We have done a series of installations in Manesar, which alone has about 5000 roofs. Using this, Gurgaon by itself can bring in about 40MW of solar power in a span of two months. Godrej itself is setting up the systems to generate 5MW of power on our own rooftops," says Nitesh Madan, manager - north PIRE, Godrej Electricals & Electronics.

Transformers play a big role in transmission losses. "What is happening today is that, low-quality equipment are used because there is no prior inspection being done by the client. The second is the project-management-execution capability and quality that plays an important role. Even if good equipment is used but cable execution was not properly done, it will cause an increase in losses," according to Simarpreet Singh, head strategy, Hartek Power Pvt Ltd. Quality should be maintained by having inhouse testing facilities that are accredited by a third party like National Accreditation Board for Testing and Calibration Laboratories (NABL).